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Information Theory for Data Science and Machine Learning

Problem 1 Chebyshev's Inequality for Binomial Distribution

Chebyshev's Inequality: $P(|X - E[X]| \geq \varepsilon) \leq \frac{Var(X)}{\varepsilon^2}$

$S_N = \sum_{i=1}^{N} X_i \sim B(N, 1/2)$ ← fair coin

$E[S_N] = E[\sum_{i=1}^{N} X_i] = \sum_{i=1}^{N} E[X_i] = N(1 \cdot p(X_i=1) + 0 \cdot p(X_i=0)) = Np = \frac{N}{2}$

$Var[S_N] = Var[\sum_{i=1}^{N} X_i] \overset{\text{independent}}{=} \sum_{i=1}^{N} Var[X_i] = N(E[X_i^2] - E[X_i]^2)$

$= N((1^2 \cdot p(X_i=1) + 0^2 \cdot p(X_i=0)) - p^2) = N(p - p^2)$

$= Np(1-p) = \frac{N}{4}$

$P(|S_N - E[S_N]| \geq \frac{N}{4}) \leq \frac{Var(S_N)}{(\frac{N}{4})^2}$

$\Leftrightarrow P(|S_N - \frac{N}{2}| \geq \frac{N}{4}) \leq \frac{\frac{N}{4}}{\frac{N^2}{16}} = \frac{4}{N}$

$(|S_N - \frac{N}{2}| \geq \frac{N}{4}) = (S_N - \frac{N}{2} \geq \frac{N}{4}) \sqcup (S_N - \frac{N}{2} \leq -\frac{N}{4})$

$P(S_N \geq \frac{3}{4}N) = P(S_N - \frac{N}{2} \geq \frac{N}{4}) = P(|S_N - \frac{N}{2}| \geq \frac{N}{4}) - \underbrace{P(S_N - \frac{N}{2} \leq -\frac{N}{4})}_{\geq 0}$

$\leq P(|S_N - \frac{N}{2}| \geq \frac{N}{4}) \leq \frac{4}{N} \rightarrow k = 4$

Problem 2 Tails of the Normal Distribution $Z \sim \mathcal{N}(0,1) \quad \forall r > 0$

- $P(Z \geq r) = \frac{1}{\sqrt{2\pi}} \int_r^\infty 1 \cdot e^{-\frac{x^2}{2}} dx \leq \frac{1}{2\pi} \int_r^\infty \frac{x}{r} e^{-x^2/2} dx \quad (x \geq r \Rightarrow \frac{x}{r} \geq 1)$ $(r>0)$

$\leq \frac{1}{r} \cdot \frac{1}{\sqrt{2\pi}} \int_r^\infty x \cdot e^{-x^2/2} dt = \frac{1}{r} \cdot \frac{1}{\sqrt{2\pi}} [-e^{-x^2/2}]_r^\infty = \frac{1}{r} \cdot \frac{1}{\sqrt{2\pi}} (0 - (-e^{-r^2/2}))$

$P(Z \geq r) \leq \frac{1}{r} \cdot \frac{1}{\sqrt{2\pi}} e^{-r^2/2}$

- $P(Z \geq r) = \frac{1}{\sqrt{2\pi}} \int_r^\infty e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}} \int_r^\infty \frac{1}{x} \cdot x e^{-x^2/2} dx$

$$P(Z \geq r) = \frac{1}{\sqrt{2\pi}} \int_r^\infty \frac{1}{x} \cdot x \cdot e^{-x^2/2}\, dx = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} \cdot \left(-e^{-x^2/2}\right)\Big|_r^\infty - \int_r^\infty \frac{-1}{x^2} \left(-e^{-x^2/2}\right) dx \right)$$

$$= \frac{1}{\sqrt{2\pi}} \left(\left(0 - \left(-\frac{1}{r} e^{-r^2/2}\right)\right) - \int_r^\infty \frac{1}{x^3} \cdot x\, e^{-x^2/2}\, dx \right) \left(\leq \frac{1}{\sqrt{2\pi}} \frac{1}{r} e^{-r^2/2} - 0 \right)$$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{r} e^{-r^2/2} - \left(\frac{1}{x^3} \cdot \left(-e^{-x^2/2}\right)\Big|_r^\infty - \int_r^\infty \frac{-3}{x^4} \cdot \left(-e^{-x^2/2}\right) dx \right) \right)$$

$$= \frac{1}{\sqrt{2\pi}} e^{-r^2/2} \left(\frac{1}{r} - \frac{1}{r^3} \right) + \underbrace{\int_r^\infty \frac{3}{x^4} e^{-x^2/2}\, dx}_{\geq 0}$$

$$\geq \frac{1}{\sqrt{2\pi}} e^{-r^2/2} \left(\frac{1}{r} - \frac{1}{r^3} \right)$$

Finally $\left(\frac{1}{r} - \frac{1}{r^3}\right) \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-r^2/2} \leq P(Z \geq r) \leq \frac{1}{r} \cdot \frac{1}{\sqrt{2\pi}} e^{-r^2/2}$

Problem 3 Central Limit Theorem and Tails of the Normal Distribution

$$P\left(S_N \geq \frac{3}{4}N\right) = P\left(S_N - E[S_N] \geq \frac{3}{4}N - E[S_N]\right) = P\left(S_N - E[S_N] \geq \frac{3}{4}N - \frac{N}{2}\right)$$

$$= P\left(\frac{S_N - E[S_N]}{\sqrt{Var(S_N)}} \geq \frac{\frac{N}{4}}{\sqrt{Var(S_N)}} \right) = P\left(\frac{S_N - E[S_N]}{\sqrt{Var(S_N)}} \geq \underbrace{\sqrt{\frac{N}{4}}}_{= \frac{\frac{N}{4}}{\sqrt{\frac{N}{4}}}} \right)$$

Lindeberge - Lévy Central limit theorem: $\lim_{N \to \infty} P(Z_N \geq r) = P(Z \geq r) = \frac{1}{\sqrt{2\pi}} \int_r^\infty e^{-x^2/2}\, dx$

$Z \sim N(0,1)$

$$P(S_N \geq \tfrac{3}{4}N) = P\left(Z_N \geq \frac{\sqrt{N}}{2}\right) \overset{N \text{ big}}{\approx} P\left(Z \geq \frac{\sqrt{N}}{2}\right)$$

$P(Z \geq r) \leq \frac{1}{r} \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-r^2/2}$

Problem 2 Upper Bound

$$\leq \frac{1}{\frac{\sqrt{N}}{2}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{\sqrt{N}}{2}\right)^2/2}$$

N big $\Rightarrow N \geq 4 \Rightarrow \quad \geq 1$

$$\leq \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{N}{8}}$$